

Sales Tracker in Google Sheets

Project Report — Data Analytics Track Internship

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Organisation	Veda Technology
Track	Data Analytics
Task	Sales Tracker in Google Sheets
Deliverable	sales_tracker.xlsx (Google Sheets compatible)
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1. Objective

Build a lightweight sales tracker spreadsheet that automatically calculates daily, weekly and monthly totals from raw sale entries, so that a non-technical user can keep adding rows of data and see correct roll-ups without touching a single formula. The secondary objective was to practise structuring a spreadsheet for ongoing data entry and automatic aggregation.

2. Tools & Technologies

- Google Sheets / Microsoft Excel (.xlsx format, fully compatible with both)
- Spreadsheet functions: SUMIFS, EOMONTH, SUMPRODUCT, ROUND, IF
- Data Validation (dropdown list, date rule, numeric rules)
- Freeze panes, cell formatting and colour-coded input cells

3. Dataset Used

A retail sales dataset was used as the sample data for the tracker. Every row represents one sale.

Attribute	Detail
Records	10,194 sale entries
Date range	3 January 2023 to 30 December 2026 (1,242 distinct trading days)
Columns	Order Date, Category, Sales, Quantity
Categories	Office Supplies, Furniture, Technology
Total sales value	2,326,534.35

Attribute	Detail
Total quantity sold	38,654 units
Data quality	No blank values, no negative sales or quantities

4. Spreadsheet Structure

The workbook deliberately separates **data entry** from **calculation**, using two tabs:

4.1 Tab 1 — "Raw Entries" (input layer)

- Rows 1–2 carry a short instruction legend telling the user exactly what to do and what not to touch.
- Row 4 holds the headers: Order Date | Category | Sales | Quantity.
- Data starts from row 5; the header row is frozen so it stays visible while scrolling.
- Entry rows are highlighted so the user can see at a glance where new data belongs.
- No formulas exist on this tab at all — it is purely raw data.

4.2 Tab 2 — "Summary" (calculation layer)

Three independent roll-up blocks sit side by side, all driven by formulas that read from the Raw Entries tab:

Block	Columns	Rows generated	Period covered
Daily Totals	Date, Total Sales	1,242	Every date present in the data
Weekly Totals	Week Start, Week End, Total Sales	209	Mon–Sun weeks, Jan 2023 – Jan 2027
Monthly Totals	Month, Total Sales	48	Jan 2023 – Dec 2026

5. Formulas Used

Daily total — matches a single exact date:

```
=SUMIFS('Raw Entries'!$C$5:$C$10238, 'Raw Entries'!$A$5:$A$10238, A6)
```

Weekly total — sums everything between the week-start and week-end dates:

```
=SUMIFS('Raw Entries'!$C$5:$C$10238, 'Raw Entries'!$A$5:$A$10238, ">="&D6, 'Raw Entries'!$A$5:$A$10238, "<="&E6)
```

Monthly total — EOMONTH finds the last day of the month automatically, so month length and leap years are handled without hardcoding:

```
=SUMIFS('Raw Entries'!$C$5:$C$10238, 'Raw Entries'!$A$5:$A$10238, ">="&H6, 'Raw Entries'!$A$5:$A$10238, "<="&EOMONTH(H6,0))
```

All ranges are absolute (\$) so a formula can be copied down without the reference drifting, and every range is extended beyond the current last row so newly added entries are picked up automatically.

6. Data Validation (preventing bad entries)

Column	Rule applied	What it prevents
Order Date	Must be a valid date on or after 01-01-2020	Text like "yesterday", typos such as 12/45/2025

Column	Rule applied	What it prevents
Category	Dropdown list: Office Supplies, Furniture, Technology	Spelling variants ("Tech", "technology ") that break SUMIFS matching
Sales	Decimal number, 0 or greater	Negative values and text entered in a numeric column
Quantity	Whole number, 0 or greater	Fractional or negative quantities

7. Verification Against a Manual Check

The task required totals to be verified against a manual check, so a verification block was added at the bottom of the Summary tab. Each check recomputes the same figure using an **independent formula** — SUMPRODUCT instead of SUMIFS — and compares the two with a Match / Mismatch flag. If the logic were wrong, the two methods would disagree.

Check	Period	SUMIFS result	SUMPRODUCT cross-check	Status
Sample day	03-02-2025	866.40	866.40	Match
Sample week	30-12-2024 to 05-01-2025	8,459.56	8,459.56	Match
Sample month	January 2025	18,830.33	18,830.33	Match

7.1 Full reconciliation test

Beyond the spot-checks, all three roll-up blocks were summed and compared with the grand total of the raw data. If any date range overlapped or any day were missed, these four numbers would not agree.

Source	Total sales
Sum of all raw entries (column C)	2,326,534.35
Sum of all Daily Totals	2,326,534.35
Sum of all Weekly Totals	2,326,534.35
Sum of all Monthly Totals	2,326,534.35

Result: all four figures reconcile exactly, and the workbook contains 1,508 formulas with zero formula errors.

8. Observations from the Sample Data

Metric	Value
Highest-selling category	Technology — 839,893.28 across 1,865 orders
Highest-volume category	Office Supplies — 6,128 orders, 731,893.31
Furniture	754,747.76 across 2,201 orders
Best month	November 2026 — 118,454.50
Weakest month	February 2023 — 4,519.89
Average sale value	228.23 per order
Average daily sales	1,873.22 per trading day

The pattern is intuitive: Office Supplies sells most often but at a low ticket size, while Technology sells less often at a much higher ticket size — which is why it leads on revenue despite the lowest order count.

9. Design Decisions & Interview Notes

9.1 How would you design a spreadsheet so non-technical staff can enter data safely?

- Give them exactly one place to type — a single input tab with shaded entry cells, so there is no ambiguity about where data goes.
- Constrain the input instead of trusting it: dropdowns for categories, date rules for dates, numeric rules for amounts. A wrong entry is rejected at the moment it is typed, not discovered a month later.
- Write the instructions inside the file itself, at the top of the entry tab, so the file explains itself without a separate manual.
- Freeze the header row so the user always knows which column they are in.
- Keep the summary tab read-only in practice (and protected in a shared environment) so nobody can overwrite a formula by accident.
- Build the roll-ups so they update automatically — the moment a user has to "recalculate something manually", errors start.

9.2 What is the risk of mixing raw data and formulas on the same tab?

- A user inserting, deleting or sorting rows can silently break or shift formula references.
- Formulas sitting next to data get copied down or overwritten during ordinary data entry.
- Sorting or filtering a sheet that contains both can scramble the relationship between a formula and the row it was meant to describe.
- Exporting the data to CSV or into a BI tool becomes messy, because the export carries totals and labels mixed in with actual records.
- Auditing becomes hard — there is no single clean source of truth to trace a number back to.
- Separation keeps the raw tab append-only and the summary tab derived, which is the same principle as separating a database table from a view.

10. Limitations & Future Scope

- The Daily Totals list covers the dates present in the current dataset; entries added on a brand-new date need one more row dragged down (weekly and monthly blocks already extend to January 2027 and pick such entries up automatically).
- Totals are computed on sales value; a category-wise and quantity-wise breakdown could be added as a further roll-up.
- A dashboard tab with a trend chart and top-category chart would make the summary easier to read at a glance.
- For a genuinely multi-user setup, the Summary tab should be sheet-protected so only the raw tab is editable.

11. Conclusion

The tracker meets both deliverables of the task. It provides a reusable template whose daily, weekly and monthly totals are driven entirely by formulas, and it ships with 10,194 rows of sample data

whose totals have been verified twice — by an independent SUMPRODUCT cross-check on sample periods, and by a full reconciliation of all three roll-ups against the raw grand total. The structural choices — a separate entry tab, validated inputs and an in-file legend — make the file safe for a non-technical person to use day after day without any spreadsheet knowledge.